

Home work on Constructors

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Question 7

Design a class **Convert** to find the date and the month from a given day number for a particular year. [10]

Example: If day number is 64 and the year is 2020, then the corresponding date would be:

March 4, 2020 i.e. $(31 + 29 + 4 = 64)$

Some of the members of the class are given below:

Class name : **Convert**

Data members/instance variables:

n	: integer to store the day number
d	: integer to store the day of the month (date)
m	: integer to store the month
y	: integer to store the year

Methods/Member functions:

Convert ()	: constructor to initialize the data members with legal initial values
void accept()	: to accept the day number and the year
void day_to_date()	: converts the day number to its corresponding date for a particular year and stores the date in 'd' and the month in 'm'
void display()	: displays the month name, date and year

Specify the class **Convert** giving details of the **constructor()**, **void accept()**, **void day_to_date()** and **void display()**. Define a **main()** function to create an object and call the functions accordingly to enable the task.

Question 8

Design a class **BinSearch** to search for a particular value in an array. [10]

Some of the members of the class are given below:

Class name : **BinSearch**

Data members/instance variables:

arr[]	: to store integer elements
n	: integer to store the size of the array

Member functions/methods:

BinSearch(int nn)	: parameterized constructor to initialize n=nn
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void fillarray()	:	to enter elements in the array
void sort()	:	sorts the array elements in ascending order using any standard sorting technique
int bin_search(int l,int u,int v)	:	searches for the value 'v' using binary search and recursive technique and returns its location if found otherwise returns -1

Define the class **BinSearch** giving details of the **constructor()**, **void fillarray()**, **void sort()** and **int bin_search(int,int,int)**. Define the **main()** function to create an object and call the functions accordingly to enable the task.

Question 9

A class **Mix** has been defined to mix two words, character by character, in the following **[10]** manner:

The first character of the first word is followed by the first character of the second word and so on. If the words are of different length, the remaining characters of the longer word are put at the end.

Example: If the First word is "JUMP" and the second word is "STROLL", then the required word will be "JSUTMRPOLL"

Some of the members of the class are given below:

Class name : **Mix**

Data member/instance variable:

wrđ	:	to store a word
len	:	to store the length of the word

Member functions/methods:

Mix()	:	default constructor to initialize the data members with legal initial values
void feedword()	:	to accept the word in UPPER case
void mix_word(Mix P, Mix Q)	:	mixes the words of objects P and Q as stated above and stores the resultant word in the current object
void display()	:	displays the word

Specify the class **Mix** giving the details of the **constructor()**, **void feedword()**, **void mix_word(Mix, Mix)** and **void display()**. Define the **main()** function to create objects and call the functions accordingly to enable the task.